PERHAPS TIME IS THE GREATEST MYSTERY

1. Aristotle, *Metaphysics* I.2.982.

2. The layering of the notion of time is discussed in depth, for example, in J. T. Fraser, *Of Time, Passion, and Knowledge* (New York: Braziller, 1975).

3. The philosopher Mauro Dorato has insisted on the necessity to render the elementary conceptual framework of physics coherent with our experience; see Mauro Dorato, *Che cos'è il tempo?* (Rome: Carocci, 2013).

1. LOSS OF UNITY

4. This is the essence of the theory of general relativity. Albert Einstein, "Die Grundlage der algemeinen Relativitätstheorie," *Annalen der Physik* 49 (1916): 769–822.

5. In the approximation of a weak field, the metrics can be written $ds^2 = (1 + 2\phi(x))\, dt^2 - dx^2$, where $\phi(x)$ is the potential of Newton. Newtonian gravity follows from the sole modification of the temporal component of the metrics g_{oo}, that is, from the local slowing down of time. The geodesics of these metrics describe the fall of bodies: they bend toward the lowest potentiality, where time slows. (These and similar notes are for those who have some familiarity with theoretical physics.)

6. Carlo Rovelli, *Che cos'è la scienza: La rivoluzione di Anassimandro* (Milan: Mondadori, 2011); English translation, *The First Scientist: Anaximander and His Legacy* (Chicago: Westholme, 2011).

7. For example: $(t_{table} - t_{ground}) = gh/c^2 t_{ground}$ where c is the speed of light, $g = 9.8 m/s^2$ is the acceleration of Galileo, and h is the height of the table.

8. They can also be written with a single variable, t, the "temporal coordinate," but this does not indicate the time measured by a clock (determined by ds, not by dt) and may be changed arbitrarily without changing the world described. This t does not represent a physical quantity. What clocks measure is the proper time along a line of the universe γ, given by $t_\gamma = \int_\gamma \sqrt{g_{ab}(x) dx^a dx^b}$. The physical relation between this quantity and $g_{ab}(x)$ is discussed further on.

2. LOSS OF DIRECTION

9. Rainer Maria Rilke, *Duineser Elegien*, in *Sämtliche Werke*, vol. 1, I, vv. 83–85 (Frankfurt: Insel, 1955).

10. The French Revolution was an extraordinary moment of scientific vitality in which the bases of chemistry, biology, analytic mechanics, and much else were founded. The social revolution went hand in hand with the scientific one. The first revolutionary mayor of Paris was an astronomer; Lazare Carnot was a mathematician; Marat considered himself to be, above all else, a physicist. Antoine Lavoisier was active in politics. The mathematician Joseph-Louis Lagrange was honored by the different governments that succeeded each other in that tormented and magnificent moment